Sachin Ashok

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linkedin google scholar github

Research Interests

Networked Systems; with an emphasis on *Congestion Control* and *Network Simulation*

Education

2021 - **PhD.** in Computer Science
University of Illinois Urbana-Champaign
Advisors: **Dr. Radhika Mittal**, **Dr. P. Brighten Godfrey**

2014 – 2018 **B.Tech.** in Computer Science and Engineering
National Institute of Technology, Trichy

Research Experience

2018 - 2020 **Microsoft Research, India** **Research Fellow**
Mentors: **Dr. Venkat Padmanabhan**, **Dr. Johannes Gehrke**

- **iBox: Internet In A Box**
  - Led the effort to design a data-informed network simulator for recreating the behaviour of paths in a target network such as the Internet.
  - Designed parameterised network models which included a cross-traffic model which was critical for recreating network behaviour in challenging settings like cellular networks.
  - Integrated iBox into NetEm and Skype's in-house simulator and made publicly available the derived iBoxNet profiles for network researchers.
- **R3Net: Reinforcement Learning for bandwidth control**
  - Coordinated on an effort across MSR India, MSR Asia & Microsoft Redmond to use reinforcement learning for bandwidth estimation & congestion control.
  - Developed realistic network environments to train RL agents.

2017 **Samsung Research, India** **Research Intern**
Mentors: Dr. Pratibha Moogi, Mr. Karthikeyan Somanathan

- **Object localization**
  - Developed an end to end system to leverage deep learning models using custom convolutional neural networks for localization of objects in a given image.
  - Entire system pipeline (image capture to model inference) was pushed to production for Samsung mobile devices.

Publications

- **iBox: Internet in a Box**
  Sachin Ashok, Sai Surya Duvvuri, Nagarajan Natarajan, Venkat Padmanabhan, Sundararajan Sellamanickam, Johannes Gehrke
  In *Proceedings of the 19th ACM Workshop on Hot Topics in Networks, HotNets '20.*
- **Reinforcement Learning for bandwidth estimation and congestion control in real-time communications**
  Joyce Fang, Martin Ellis, Bin Li, Siyao Liu, Yasaman Hosseinkashi, Michael Revow, Albert Sadovnikov, Ziyuan Liu, Peng Cheng, **Sachin Ashok**, David Zhao, Ross Cutler, Yan Lu, Johannes Gehrke
  MLForSystems, NeurIPS 2019.

Industry Experience

2016 **Mozilla Organization** **Intern**

Mentors: **Dr. Franziskus Kiefer**, **Mr. Tim Taubert**

- **Blake2 support for the Network Security Services library**
  - Implemented a Blake2 (a cryptographic hash function) module for Mozilla's NSS library.
  - The NSS library is used by Google, RedHat, and Mozilla products like Thunderbird.

Honors

- Invited for demo at Microsoft TechFest, 2020.
- Among 45 students chosen worldwide for Mozilla Winter of Security.
- Among top 100 students chosen in India by Indian Institute of Science (IISc) for Summer School.
- Among top 25 students chosen in India for Research Programme by Institute of Mathematical Sciences (IMSc).
- Finalist for Microsoft Build The Shield, a national level contest organized by Microsoft.
- Finalist in Pragyan CTF, an international level security contest held by NIT Trichy.

Selected Projects

2018 **Code Character** **[Github]**

Arena for AI bot contest

- Developed the system architecture for a tech contest where contestants write `C++` code through various APIs exposed to them to control troops and fight the computer AI in a real-time strategy game.
- Designed an efficient communication scheme to enable real-time transfer of serialized user actions to the back-end using `Protobuf`.
- Event received registrations from over 74 countries and handled more than 1000 concurrent users at peak load.

2018 **Auto-Solver** **[Github]**

AI for solving Yinsh (a board game)

- Developed an AI using `Negascout` and `Monte Carlo Tree Search` to play Yinsh.
- Designed a game simulator using `bitboards` (a specialized bit-array data structure) to speed up state space exploration by 1.5x.

Talks

2020 iBox: Internet in a Box

HotNets 2020, Virtual Event

Skills

Languages: `C++, C, Python`
Frameworks: `ns-2/ns-3, MongoDB, Protobuf, gRPC, CMake`